Lecture 35: An introduction to multi-level models

- Some GLMs we have/haven't dealt with
- What multi-levels are

Common GLMs

we have encountered

- Model counts: never convert them to proportions before the analysis, because it destroys the information about sample size.
- Difficulty arises from the different scale the parameters live in compared to the outcome variable they depict; typically log-odds for logistic/binomial regressions and log-rate for Poisson regression.
- Computing implied predictions is even more important than wit Gaussian regressions.
- Also important to look at simulating prior predictive distributions.

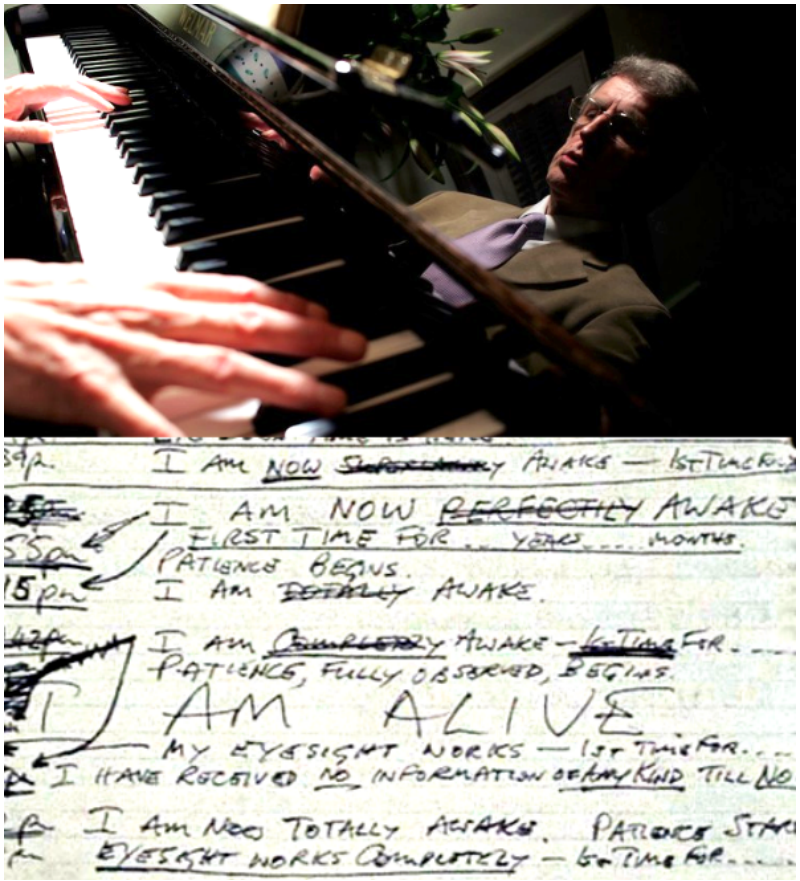
(Slightly less) common GLMs

we haven't encountered (in this course), see Chapter 12 in the SR book

- But feel free to study them in the SR book.
- Over-dispersed counts; we need to extend the binomial and Poisson models.
- Zero-inflated or zero-augmented models, which mix a binary event with an ordinary GLM likelihood (like Poisson or Binomial, but why not a continuous one like Gaussian).
- Ordered categorical models (categories with fixed ordering), where we merge a categorical likelihood function with special "cumulative" link functions.

Anterograde amnesia

- Musicologist and conductor Clive Wearing
- Lost parts of prefrontal and hippocampus
- Can still play the piano
- Can't remember he played it 5 minutes ago



Models with memory

- Models we encounter so far do not share the information from one *cluster* in the data -individual, group, location- to another if the cluster is accounted for, because we included it as a fixed effect.
- These models are programmed for amnesia. Not good for learning about the world.
- Improve and relax the independence assumption when learning from separate portions of the same data.
- Another extreme would be to ignore the cluster information and treat all portions as if they were the same.
- Instead we are going to learn simultaneously about each cluster, while learning about the “population” of clusters.
- Doing both estimation task concurrently allows to transfer information across clusters and this improves accuracy: value in remembering.
- Café example...

Multilevel models

- Two “extreme” situations for the cafés
  1. Population highly variable between clusters $\rightarrow$ prior is almost flat and uninformative $\rightarrow$ observations at any café do very little to estimate at another one.
  2. Population contains little variation between clusters $\rightarrow$ prior narrow and highly informative $\rightarrow$ observations at any café will have a big impact on estimates at any other one.
- Multilevel models will exactly do that
  - Remember features of each cluster in the data as they learn about all of the clusters
  - Depending upon the variation among clusters, which is learned from the data as well, the model pools information across clusters.
  - Pooling tends to improve estimates about each cluster. And make it faster.
- Other names: hierarchical, mixed effects (fixed & random effects). Demand mathematical or algorithmic definition.

Benefits of multilevel models

1. Improved estimates for repeat sampling; when >1 observation arises from the same individual, location or time, traditional single-level models either maximally underfit or overfit the data.
2. Improved estimates for imbalance in sampling; when some individuals/locations/times are sampled more than others, multilevel models automatically cope with different uncertainty across the clusters. Avoid domination of over-sampled clusters in inference.
3. Estimates of variation within each individual or other groups within the data explicitly.
4. Avoid averaging, retain variation; danger to average repeated measures (variation removed; which average?): it can create false confidence and introduce arbitrary data transformations.

Multilevel models are fair to the data and to its structure.

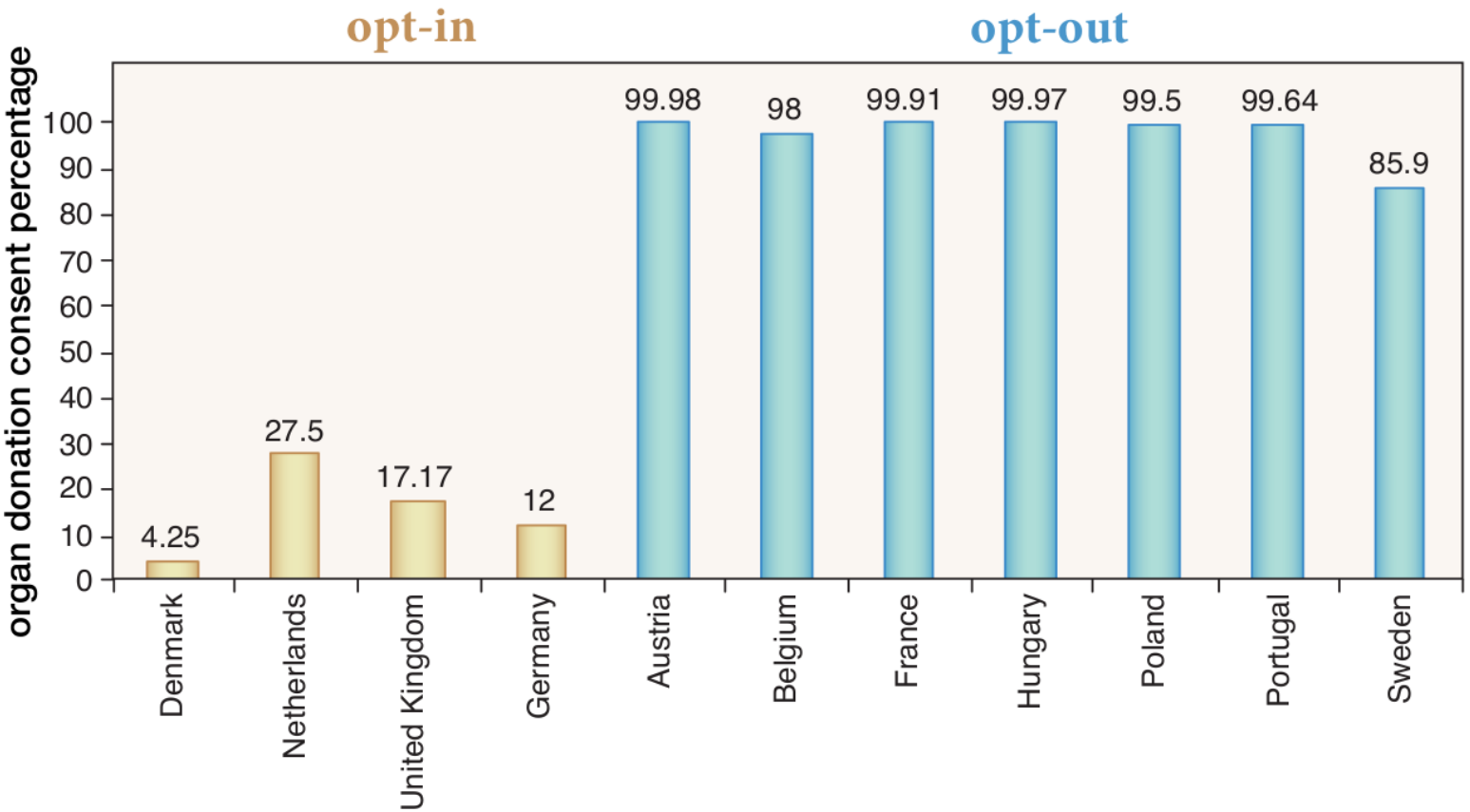
Costs of multilevel models

1. Make new assumptions, e.g. to define distributions for the cluster characteristics; entropy to the rescue.
2. New estimation challenges; MCMC estimation.
3. Not straightforward to understand; another level of complexity in predictions at different levels and parameter interpretation. We may focus on one or a few of these levels. Model comparison (e.g. WAIC) is more subtle. Basic logic is the same, need to decide which parameters to focus on.

Once the new strategy/design is learnt, you can use it as a default at least in regression scenarios: better to see it's not necessary than overlook it.
Can incorporate measurement errors or modelling missing data (see Chapter 15, SR book).

Organ donation across countries

Default or opt-in



Goals

Turtles all the way down

- Introduce multilevel models
- How *shrinkage* and *pooling* work
- Why they produce better estimates
- How to implement it with `ulam()`
- Methods for plotting and comparing
- Open to more options



Multilevel models

- Usual use is to model clustering
  - Classrooms within schools
  - Students within classrooms
  - Grades within students
  - Questions within exams
- Repeat measures of units
- Imbalance in sampling
- “pseudo-replication”



Multilevel models

Examples from previous studies

- !Kung individuals in families
- Primate species in clades
- Nations in continents
- Applicants in departments



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Example: Tadpole predation

```
library(rethinking)
data(reedfrogs)
d <- reedfrogs
```

- Outcome: number of surviving tadpoles (after eggs suspended)
- Different densities/sizes
- With/without predators
- Focus on variation across tanks
- Can fit two basic models:
  1. Dummy variable for each tank
  2. Multilevel model with *varying intercept* by tank



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Regularised intercepts

number surviving, tank i

$$S_i \sim \text{Binomial}(N_i, p_i)$$
$$\text{logit}(p_i) = \alpha_{\text{TANK}[i]}$$
$$\alpha_j \sim \text{Normal}(0, 1.5)$$

regularizing prior

Regularised intercepts

R code 13.2

```
# make the tank cluster variable
d$tank <- 1:nrow(d)

dat <- list(
    S = d$surv,
    N = d$density,
    tank = d$tank )

# approximate posterior
m13.1 <- ulam(
    alist(
        S ~ dbinom( N , p ) ,
        logit(p) <- a[tank] ,
        a[tank] ~ dnorm( 0 , 1.5 )
    ), data=dat , chains=4 , log_lik=TRUE )
```

$$S_i \sim \text{Binomial}(N_i, p_i)$$
$$\text{logit}(p_i) = \alpha_{\text{TANK}[i]}$$
$$\alpha_j \sim \text{Normal}(0, 1.5)$$

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Adaptive regularisation

$$
\begin{aligned}
S_i &\sim \text{Binomial}(N_i, p_i) \\
\text{logit}(p_i) &= \alpha_{\text{TANK}[i]} \\
\textit{varying intercepts} \longrightarrow \alpha_j &\sim \text{Normal}(\bar{\alpha}, \sigma) \\
\bar{\alpha} &\sim \text{Normal}(0, 1.5) \\
\sigma &\sim \text{Exponential}(1)
\end{aligned}
$$

Terminology

- *Varying intercepts* are also called *random intercepts*
- Neither terms makes much sense
  - "random"? Sometimes associated with research design, but design is irrelevant
  - Ordinary dummy variables also "vary" across clusters
- Distinctive because individual intercepts learn from one another
  - *mnestic*: opposite of amnestic
- Now, we learn the priors...from the data!
- The prior has a mean $\overline{\alpha}$ and a variance σ
  - Both these mean and variance have priors (priors for a prior) and associated *hyperparameters*.

Summary: survival across tanks has a distribution. This distribution is the prior for each tank. This distribution needs its own priors.

Adaptive regularisation

$$
\begin{aligned}
S_i &\sim \text{Binomial}(N_i, p_i) \\
\text{logit}(p_i) &= \alpha_{\text{TANK}[i]} \\
\alpha_j &\sim \text{Normal}(\bar{\alpha}, \sigma) \\
\bar{\alpha} &\sim \text{Normal}(0, 1.5) \\
\sigma &\sim \text{Exponential}(1)
\end{aligned}
$$

```
m13.2 <- ulam(
    alist(
        S ~ dbinom( N , p ) ,
        logit(p) <- a[tank] ,
        a[tank] ~ dnorm( a_bar , sigma ) ,
        a_bar ~ dnorm( 0 , 1.5 ) ,
        sigma ~ dexp( 1 )
    ), data=dat , chains=4 , log_lik=TRUE )
```

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Comparing both models

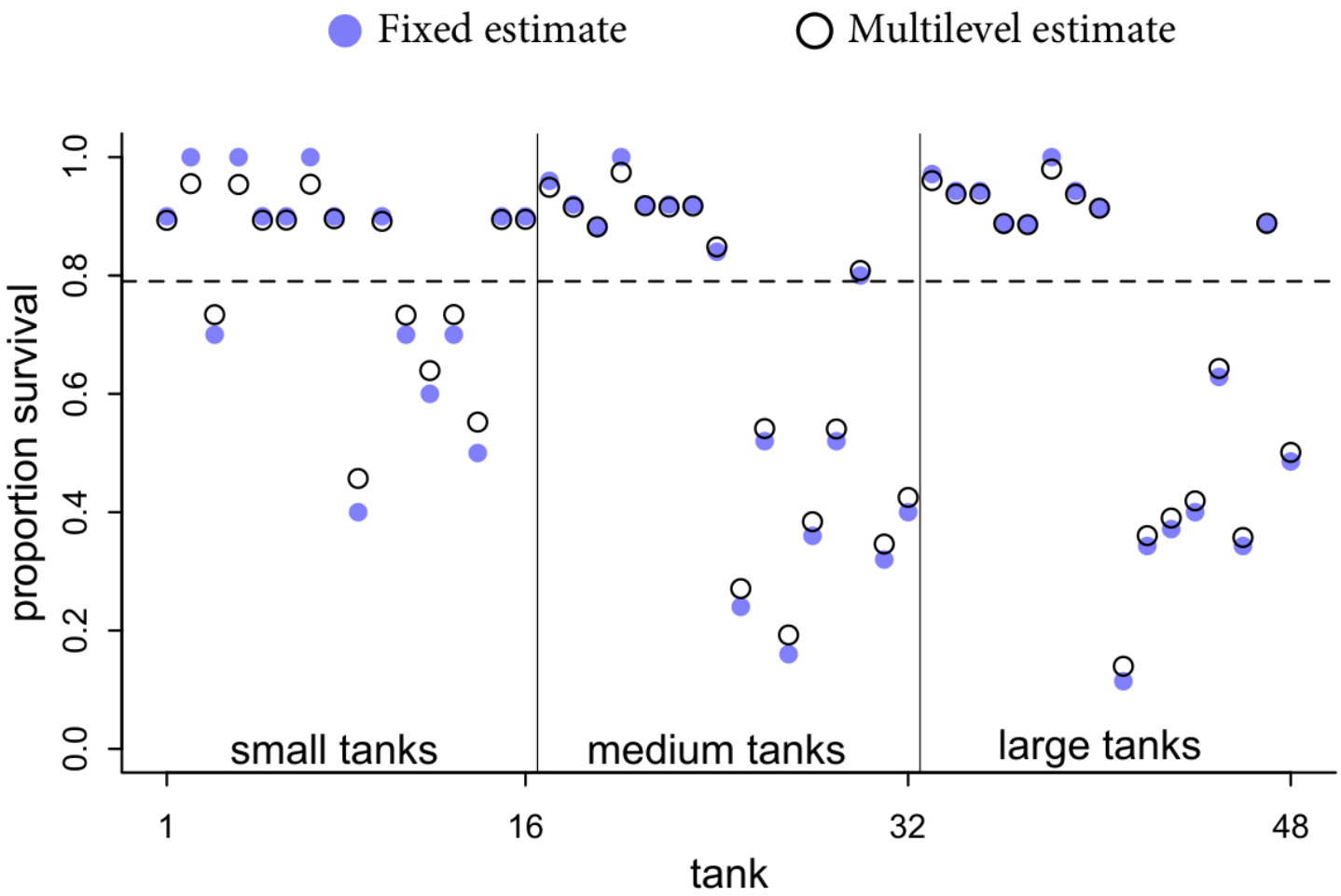
R code 13.4

```
compare( m13.1 , m13.2 )
```

```
       WAIC pWAIC dWAIC weight   SE  dSE
m13.2   202  21.7     0      1 7.35   NA
m13.1   213  24.8    11      0 4.71 3.58
```

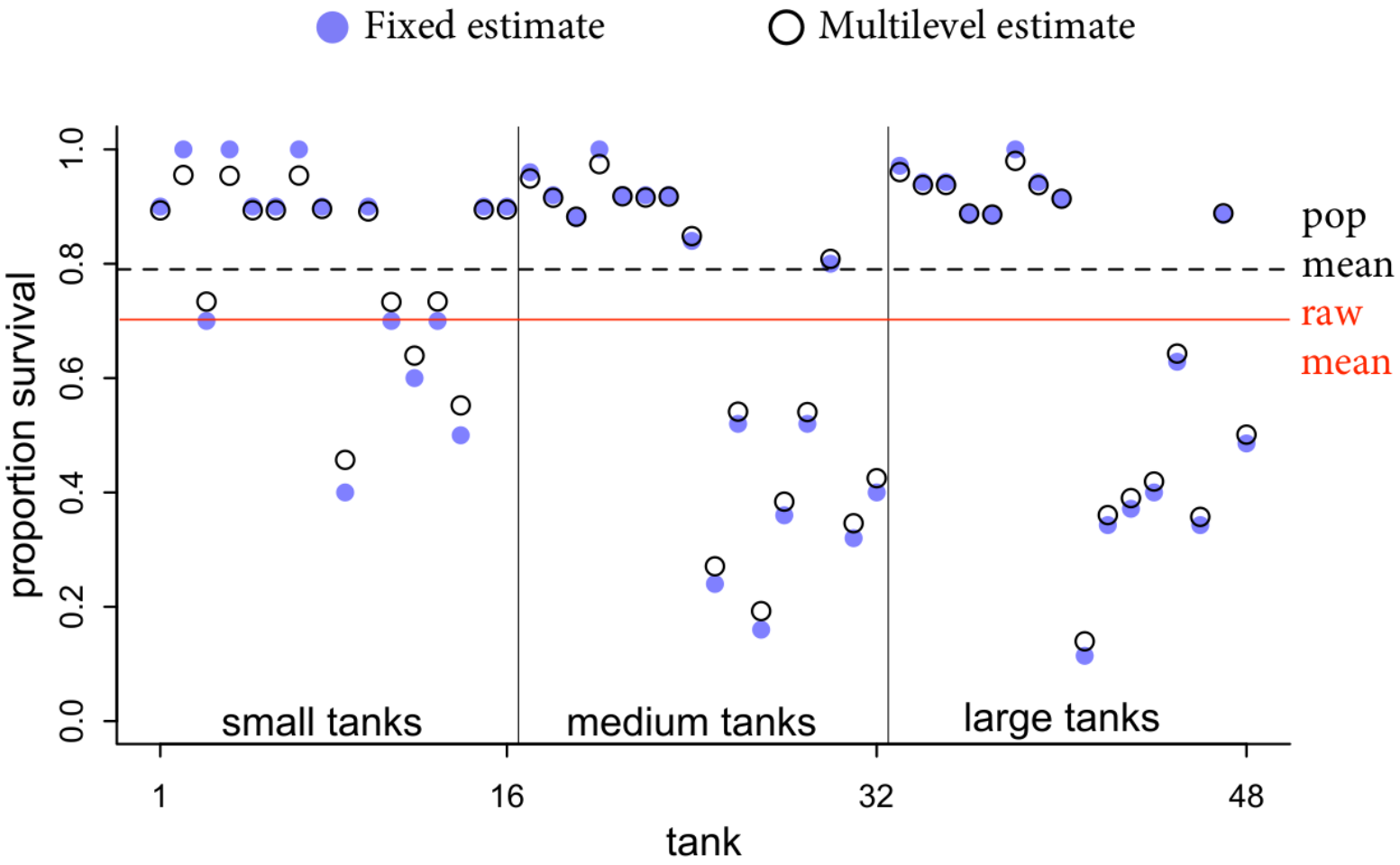
- `m13.1`: 48 parameters vs 25 effective
- `m13.2`: 50 parameters vs 22 effective
- Model with more parameters has fewer effective parameters
- Why? Ended up with stronger prior.

Posterior means from models in 48 tanks

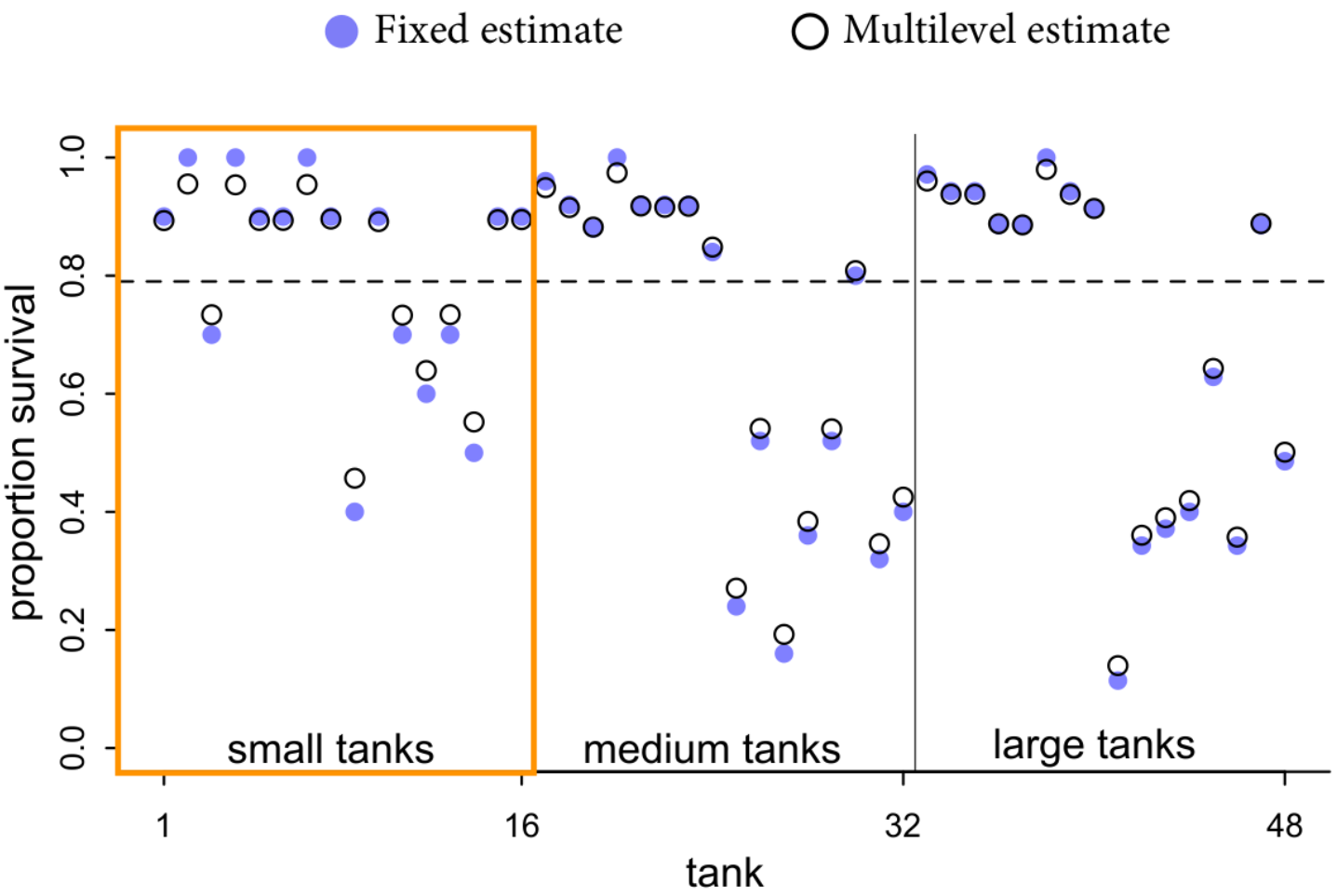


Predictions don't match observations exactly. Instead, we observe *shrinkage*.

Results from pooling of information.

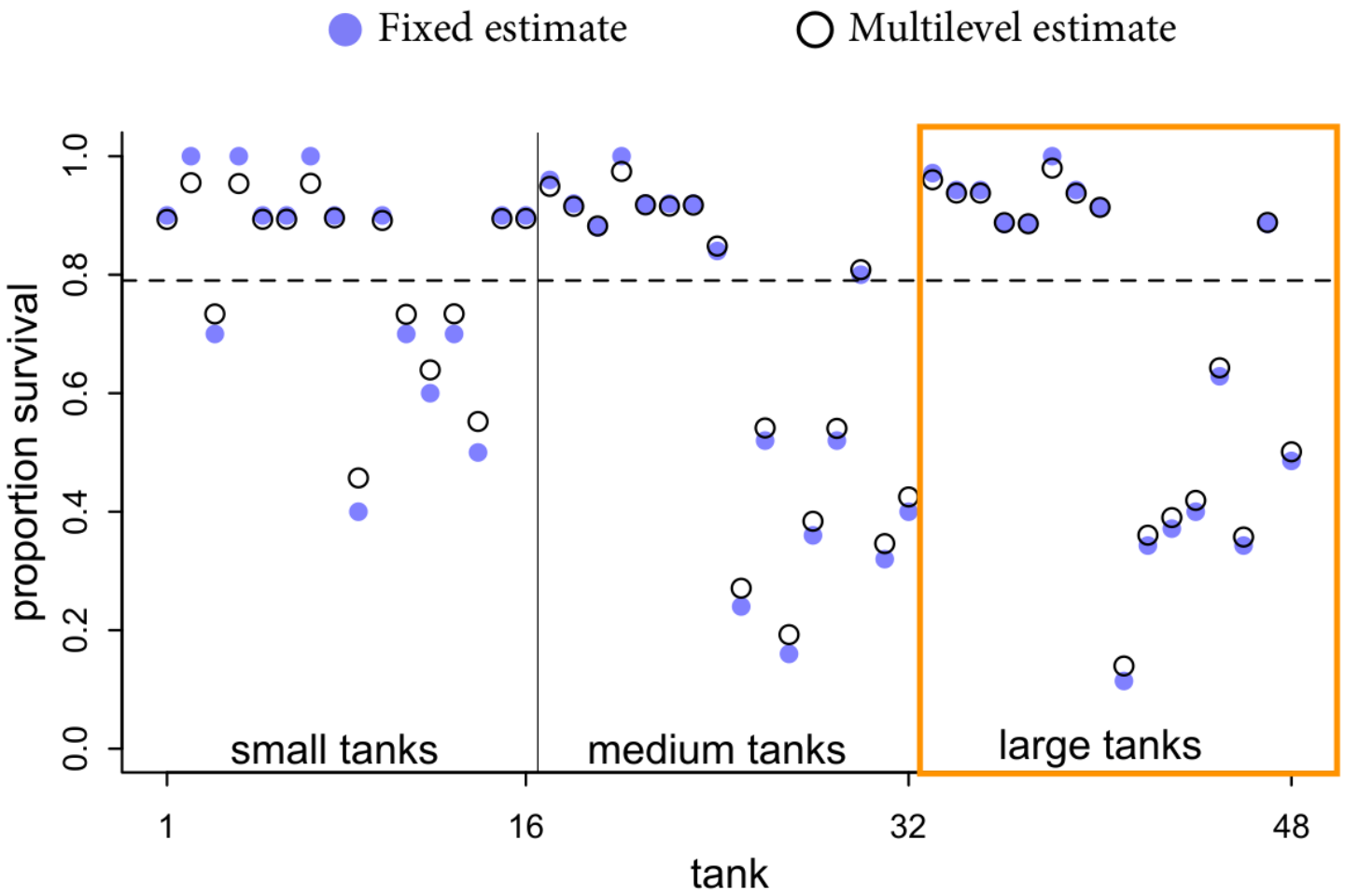


Population mean is different to the raw empirial mean. Why?
Imbalance in amount of evidence across tanks.



Small tanks $\rightarrow$ high sampling variation. More shrinkage towards the mean.
Further from the mean $\rightarrow$ more shrinkage.

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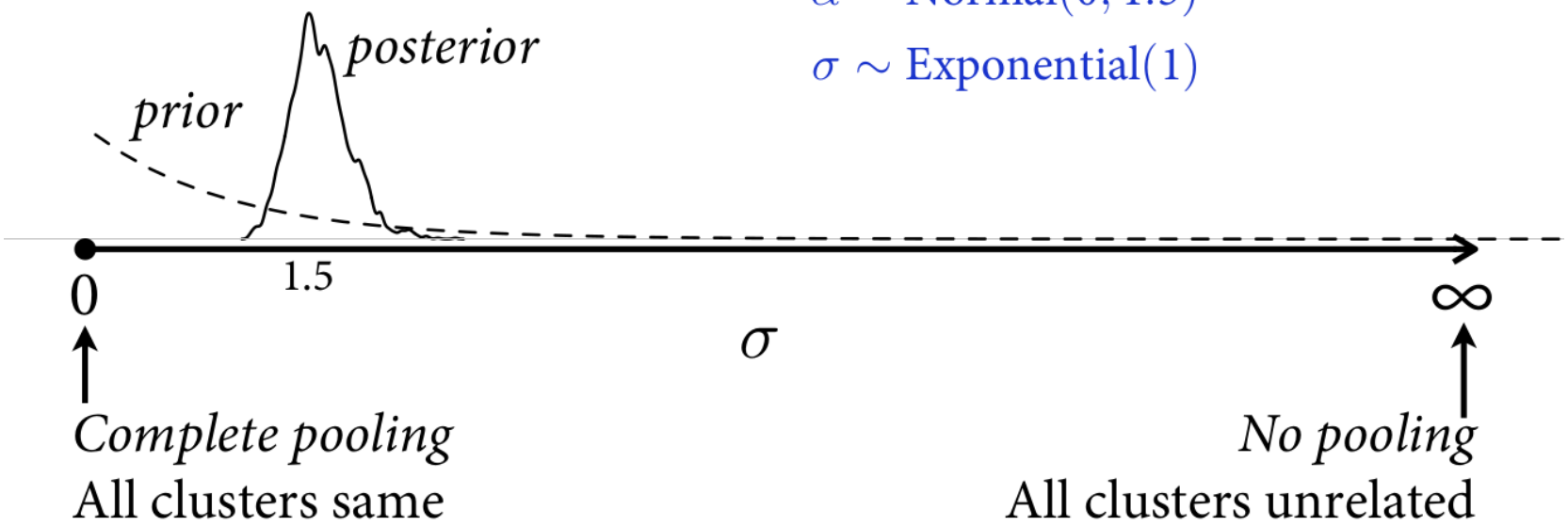


Large tanks $\rightarrow$ low sampling variation. Less shrinkage towards the mean at all distances from the mean.

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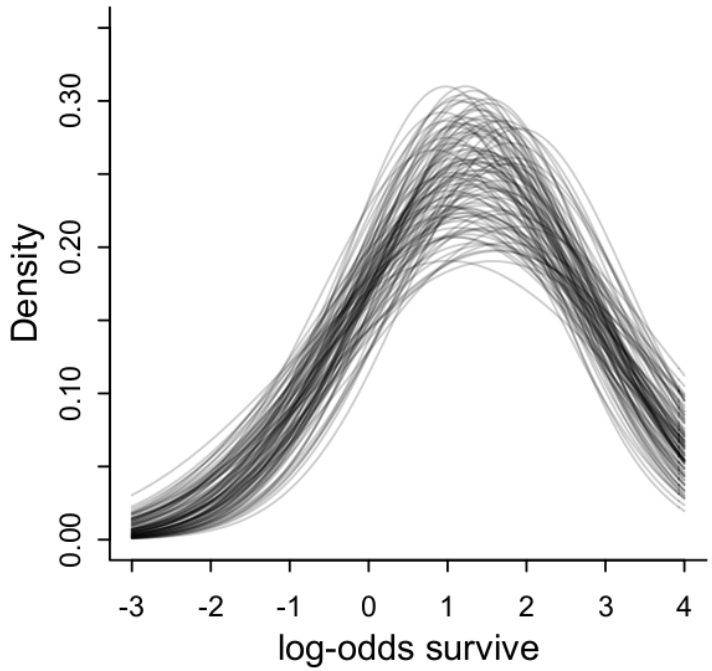
Adaptative regularisation

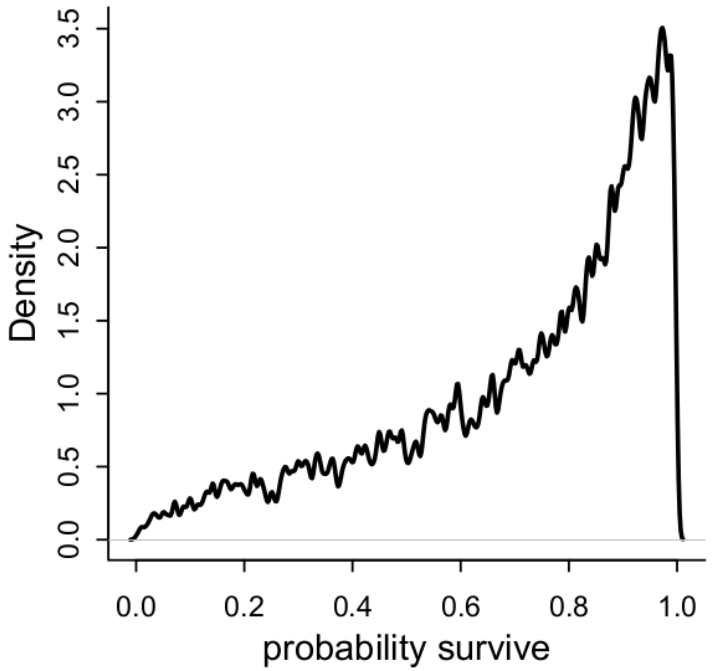
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\end{aligned}
$$



Regularising distribution

Inferred population of survival across tanks





Uncertainty about both location α and scale σ of population distribution of log-odds of survival (left).

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Varying effects - the under-/over-fitting trade-off

- Varying intercepts $=$ adaptatively regularised estimates.
- Estimates how diverse the clusters are, while estimating the features of each.
- Provides more accurate estimates of the individual cluster (tank) intercepts.
- Three perspectives
  - Complete pooling has lots of data but poor fit to individual clusters (ponds) with a unique $\alpha \rightarrow$ under-fitting (variation among ponds is 0)
  - No pooling, were we estimate different proportions from each cluster (pond) independently for predictions. Little data for each estimated value, so high error $\rightarrow$ over-fitting.
  - Partial pooling is an intermediary situation.
- Demonstration via simulations.